

SHIVIN RAJESH ANAND

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Education

University of Florida

B.S. Computer Engineering

Gainesville, FL

May 2030

Skills

Hardware & Embedded: PCB Design (KiCad), Embedded C, I2C Protocol, Circuit Design, Sensor Integration, Arduino/ESP32, Signal Debugging

FPGA & Digital Logic: Verilog, SystemVerilog, RTL Design, Vivado, Vitis, Digital Logic Design

Software & CV: Python, C++, Java, JavaScript, OpenCV, Regression Analysis, MongoDB, Git, Linux

Additional: HTML/CSS, Web Design, Robotics, Robotic Kinematics, Mechanical Design, Rapid Prototyping

Experience & Involvement

University of Florida, Dept. of ECE

Gainesville, FL

Undergraduate Research Assistant, Smart Systems Lab

August 2026 – Present

- Designing a CPU in Verilog and SystemVerilog to execute core ALU operations (addition, subtraction, multiplication) as part of an FPGA/SoC design project under PhD student Peter Forcha, using the Vivado and Vitis toolchain
- Establishing the CPU's ALU and datapath as the RTL foundation for extending it into a quantized neural-network inference accelerator, following the lab's published approach to accelerating hybrid quantized neural networks on multi-tenant FPGAs
- Targeting low-latency, near-sensor inference as the next phase of the project, building on the lab's FPGA-based near-sensor image processing accelerator research to move compute closer to the data source and reduce inference latency

IEEE Student Branch, University of Florida

Gainesville, FL

Hardware Team Member – SoutheastCon Robotics Competition

September 2026 – Present

- Designing and prototyping, on breadboard ahead of PCB fabrication in KiCad, the single custom board that will control all electrical systems on the team's autonomous competition robot for IEEE SoutheastCon's Hardware Competition
- Engineering the board's power distribution and signal architecture to support the competition's required autonomous navigation, computer vision, and movement speed-optimization subsystems
- Serving as sole electrical/PCB designer on a 10-person team, owning hardware design and validation end-to-end while coordinating with the software subteam on system integration

University of South Florida, Dept. of EE

Tampa, FL

Head Researcher, Bio-Organic Electronics Lab

April 2024 – May 2026

- Enabled continuous real-time environmental monitoring of CO₂, VOCs, temperature, and humidity by designing and fabricating a custom PCB in KiCad with I2C sensor architecture, unifying communication across a shared 3.3V bus between a BME680 sensor and Teensy microcontroller, and verifying signal integrity with a multimeter
- Reduced manual harvest requirements to zero by building a closed-loop autonomous hydroponic growth chamber (nutrient circulation, relay-switched LED control, structural housing), selecting the embedded control platform via a weighted decision-matrix analysis across Teensy, ESP32, and Arduino Uno

Middleton Robotics

Tampa, FL

President

May 2024 – May 2026

- Returned the school's FRC program to competition after years of inactivity by rebuilding the mechanical and electrical pipeline (Java on RoboRIO, C++ on VEX controllers, Git-based version control), resulting in 2 national awards and 1 World Championship award, while expanding competitive participation 30% by coordinating 9 teams across 15+ regional competitions

Florida Student Association of Mathematics

Tampa, FL

Region 4a Coordinator & Webmaster

June 2023 – May 2026

- Reduced manual communication overhead by 60% by developing and maintaining the organization's website (HTML, CSS, JavaScript, Git, Linux-based hosting), while boosting regional engagement by 50% by creating pathways for 80 Tampa Bay students into elite competitions (HMMT, PUMaC, ARML, CMIMC)

Projects

Autonomous Plant Biomass Estimation via Computer Vision | Independent Research Project (USF)

- Achieved 6.9% MAPE and R squared of 0.91 by building an OpenCV computer vision pipeline in Python (HSV conversion, Excess Green Index segmentation, morphological refinement) to autonomously extract canopy area across 6 independent growth trials
- Converted projected canopy area into fresh biomass estimates by developing a linear regression model calibrated against destructive-harvest ground truth data
- Isolated the primary sources of measurement variance by conducting systematic error analysis across trials, examining leaf overlap, edge segmentation accuracy, and calibration sensitivity
- Earned a NASA Kennedy Space Center Award by presenting this research at the Southeastern Science and Engineering Fair for its spaceflight-viability implications

Languages & Certifications

Languages: Hindi | Gujarati | Spanish (Basic)

Engineering & PM: RECF Robotics, Programming, Electrical, Mechanical, Fundamentals of Engineering | PMI Project Management Ready

Technical: Autodesk Inventor, AutoCAD & Fusion 360 (Certified User) | Linux Essentials (LE-1)